

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

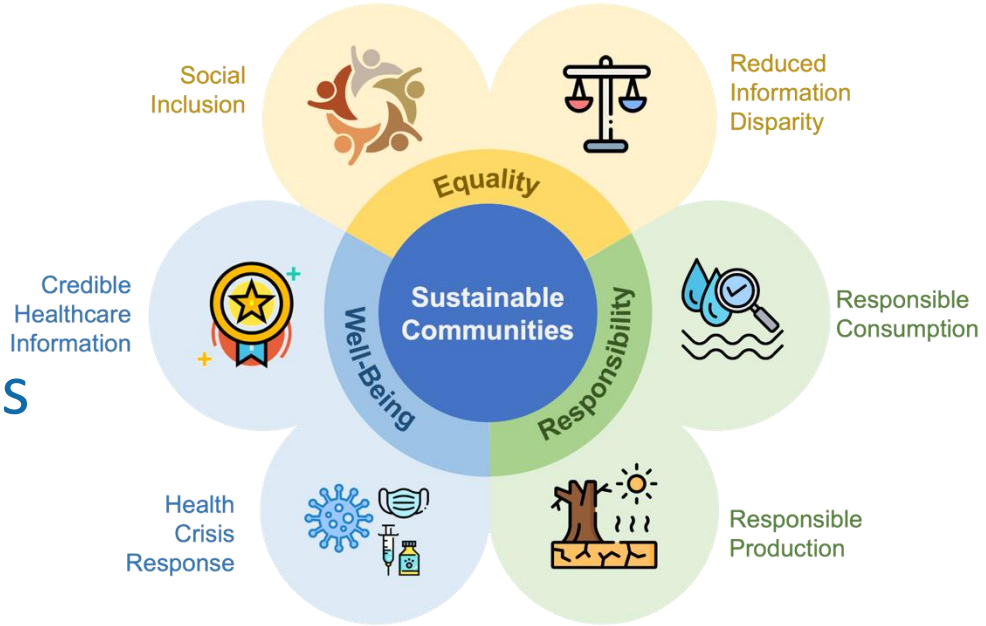
Professor Lanyu Shang

About Me

- **Lanyu Shang**
  - **Assistant Professor** in Computer Science at LMU
  - **Ph.D. in Information Science** from University of Illinois Urbana-Champaign, 2024
  - Was a Ph.D. student in **Computer Science & Engineering** at University of Notre Dame (2018-2021)
  - M.S. in **Data Science** from New York University
  - B.S. in **Applied Math** from UCLA

About My Research

- Human-Centered AI
- AI for Social Good
- Social Media Mining
- Sustainable Communities
- AI Ethics



Example of My Research

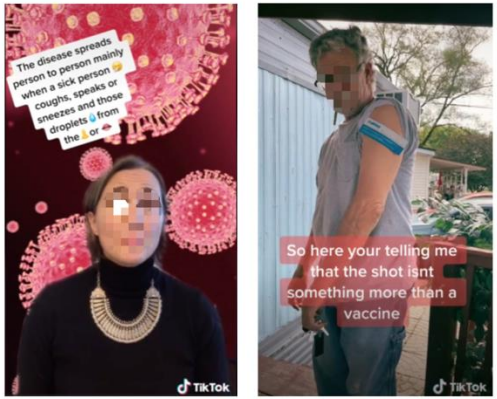
- Fuse complex human-generated **multimodal** data to identify abusive content



Offensive Meme



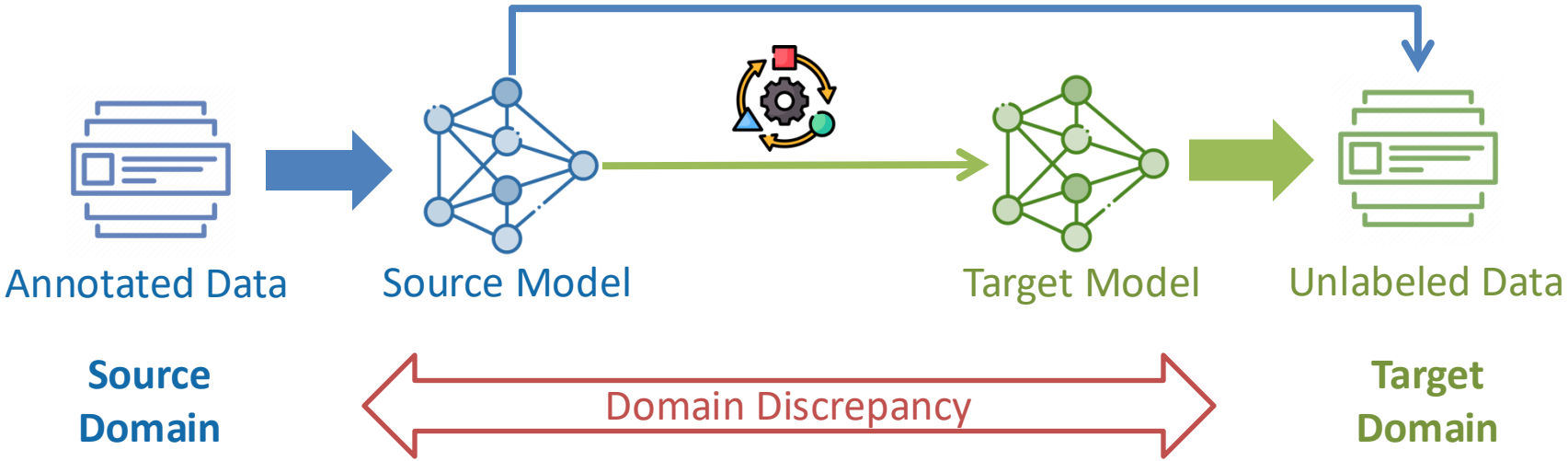
Multimodal Fake News



Misleading TikTok Video

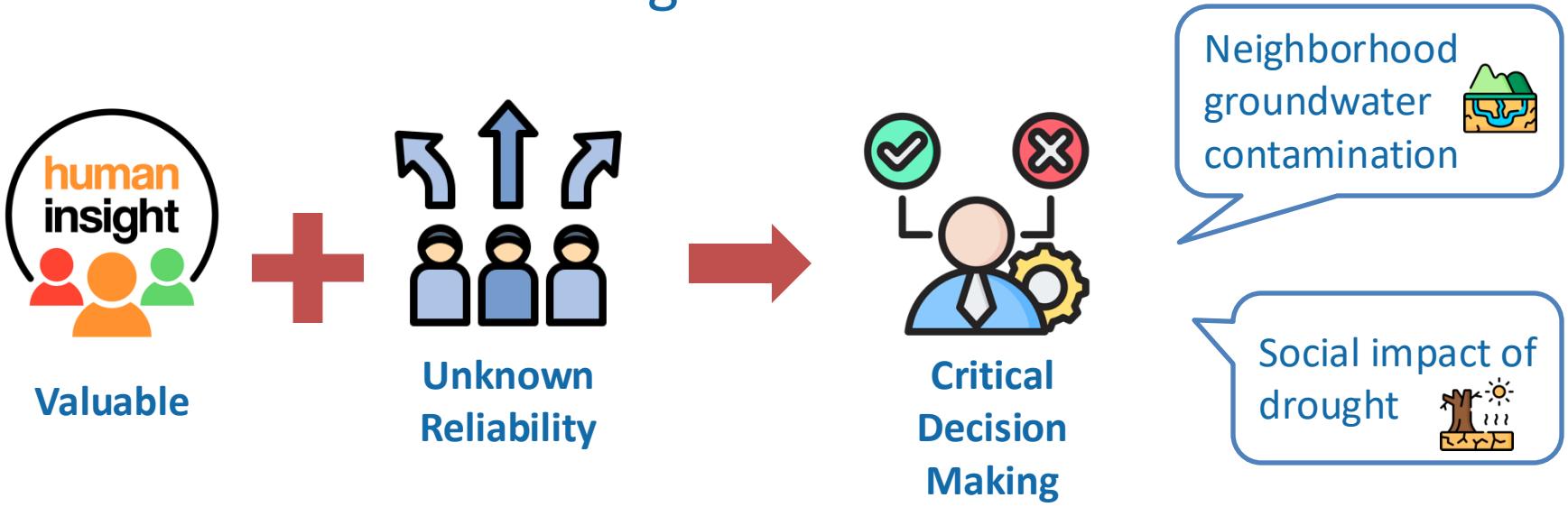
Example of My Research

- Ensure the **adaptability** of ML/AI models across discrepant domains in highly dynamic and evolving human communities



Some Examples of My Research

- Provide **trustworthy** information and insights for critical decision-making in human communities



A bit more about me …

- I wish I do not, but I make mistakes.
- Especially when it is about details
  - E.g., ∧ (AND) and ∨ (OR)
- Don’t feel afraid of speaking it out if something is not normal.
  - Statistics show that you will be right 99 out of 100 times when that happens …

About the Course

- CMSI 3300 **Section 02** - Artificial Intelligence
- Meeting Time and Location
  - Tues/Thurs 03:40 PM - 05:20 PM, **Seaver 205**
- Instructor: Lanyu Shang
  - Office Hours:
    - Tue/Thur 10 AM – Noon, Doolan 205/Zoom
    - Or by appointment
  - Email: Lanyu.Shang@lmu.edu
  - Zoom: https://lmula.zoom.us/j/86471072428

Learning Outcomes

- Be introduced to many **active research areas in AI**
  - Logical inference, probabilistic & causal reasoning
  - Fundamentals of data science, machine learning, deep learning, and generative models.
- Understand existing approaches to **classic and realistic AI problems**
  - reasoning under uncertainty, reasoning over time
  - value of information, classification

Learning Outcomes

- Practice with popular **data structures and models in AI**
  - Knowledge bases, search trees
  - Bayesian networks, decision networks
  - Classifiers and regression models
  - Ensemble methods
  - Deep neural networks

Learning Outcomes

- Be familiar with popular **AI frameworks and libraries in Python**
  - Numpy, Scikit, and PyTorch
- Grasp the **ongoing research in the field**, based on student-specific interests

Prerequisites

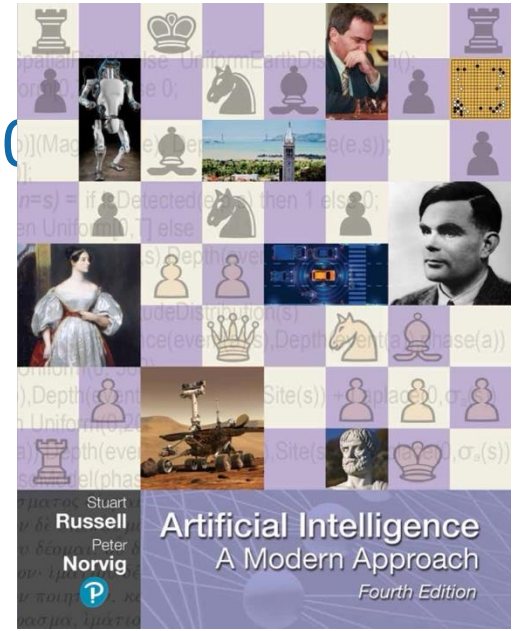
- **Python** programming language
- **Data Structures** (CMSI 2120) or its equivalent
- **Algorithms** (CMSI 2130) or its equivalent
- **Git** and/or **GitHub**

Course Logistics

- **Brightspace**
  - Announcements
  - Course materials
  - Grading/feedback
  - Group formation
  - Discussion/Q&A
- **GitHub Classroom**
  - Assignment submission
  - Coding-related feedback
  - Collaboration
  - Version control

Textbook

- **Artificial Intelligence: A Modern Approach**
  by *Stuart Russell* and *Peter Norvig*
- ISBN 9780134610993 / 9789356063570
- Optional but recommended



Attendance

- Not mandatory but highly recommended
- No need to inform the instructor of missed lecture
- Lectures are participatory and interactive
- Laptops are discouraged but note-taking are encouraged

Assignments and Grading

Classwork [25%] – Group-based 

Homework [40%] – Group-based 

Exams [35%] – Individual 

Classwork [25%]

- Small classwork assignments for in-class practice
- ~ 2 weeks (~ 7 assignments in total)
- To be finished **in groups** (except the first one)
- Lowest classwork grade will be dropped (even if you skipped it entirely).

Homework [40%]

- Larger assignments with **heavy** coding required
- ~ 3 weeks (~ 4 assignments in total)
- To be finished **in groups**
- Programming style and comments are graded as well.

Exams [35%]

- **2 Exams**: Midterm + Final
- Conceptual and mathematical (no coding)
- To be finished **individually** and **in class**
- The exams are **closed** book/notes/computer.
- You may bring **one** letter size (double-sided) cheat sheet.
- **Weighted**: exam grade = 2/3*higher one + 1/3*lower one

Final Letter Grade

- A: 93 - 100
- A-: 90 - 92
- B+: 86 - 89
- B: 83 - 85
- B-: 80 - 82
- C+: 76 - 79
- C: 73 - 75
- C-: 70 - 72
- D: 65 - 69
- F: 64 and below

Assignment Posting and Submission

- Assignments (classwork, homework, etc.) will be posted on **Brightspace** and announced **in class**.
- Will be submitted electronically on **GitHub classroom**

Deadline

- Assignments are typically due on **Fridays at 5 PM**.
- Late Policy
  - 0-24 hours: 20% penalty (the points earned * 80%)
  - 24+ hours: 0
  - No exceptions unless there are extenuating circumstances that have been discussed with and approved by the instructor prior to the deadline

Time management is also quite important!

Group

- For classwork and homework
- **Up to 3 students** in each group
- Self enroll on Brightspace by Friday **8/30 at 5PM.**
  - If you find a group OR
  - If you plan to work alone
- Otherwise, you will be **randomly assigned** to a group.

Finding a Group

- Brightspace – you will have a classwork task on this later.
- But for now,



Next…

INTRODUCTION

What is Intelligence?

- What is "intelligence," and what **qualities** do we associate with it? ->Slido

slido

Please download and install the
Slido app on all computers you use

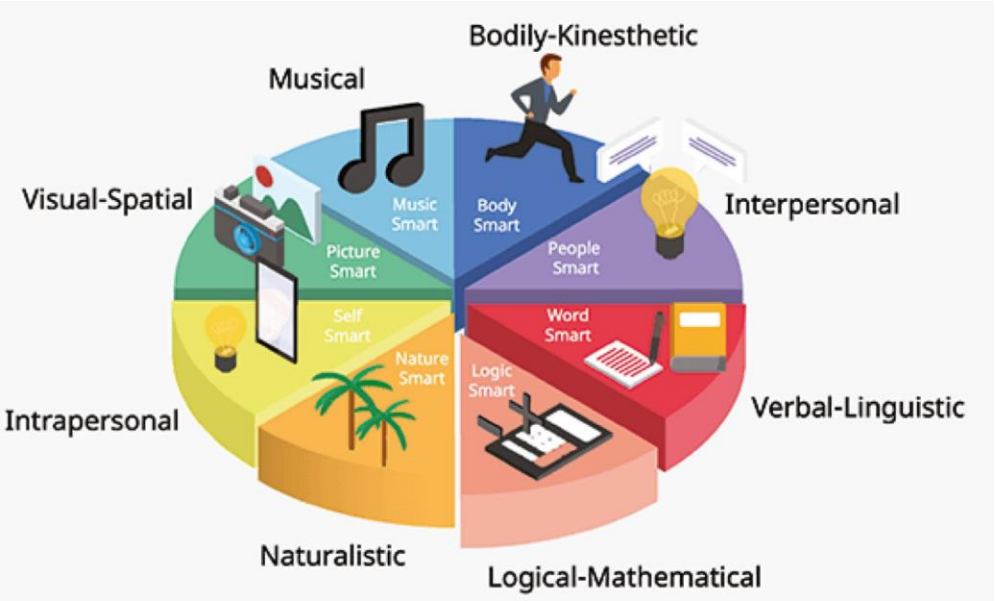


What is "intelligence," and what qualities do we associate with it?

ⓘ Start presenting to display the poll results on this slide.

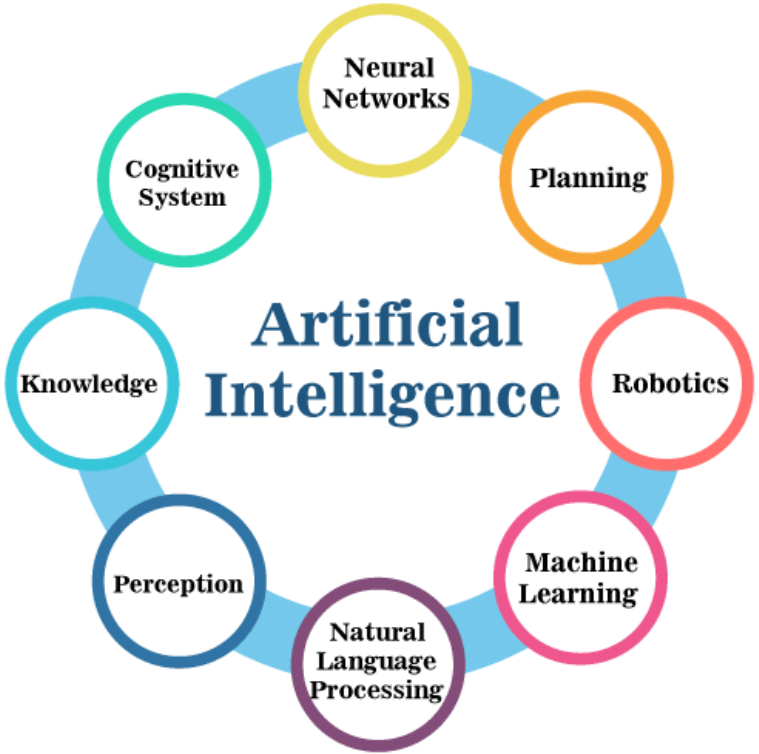
Human Intelligence

- Human intelligence refers to the **cognitive capabilities** and **mental processes** naturally possessed by humans.



Artificial Intelligence

Artificial Intelligence is the umbrella term for the study and development of **computer systems** that can perform tasks normally reserved for **human-level intellects**.



Artificial General Intelligence

Artificial General Intelligence (AGI) is the theoretical pinnacle of AI in which **a single agent** can perform ***any*** task given to it **at human level or beyond**.

Artificial Specific Intelligence

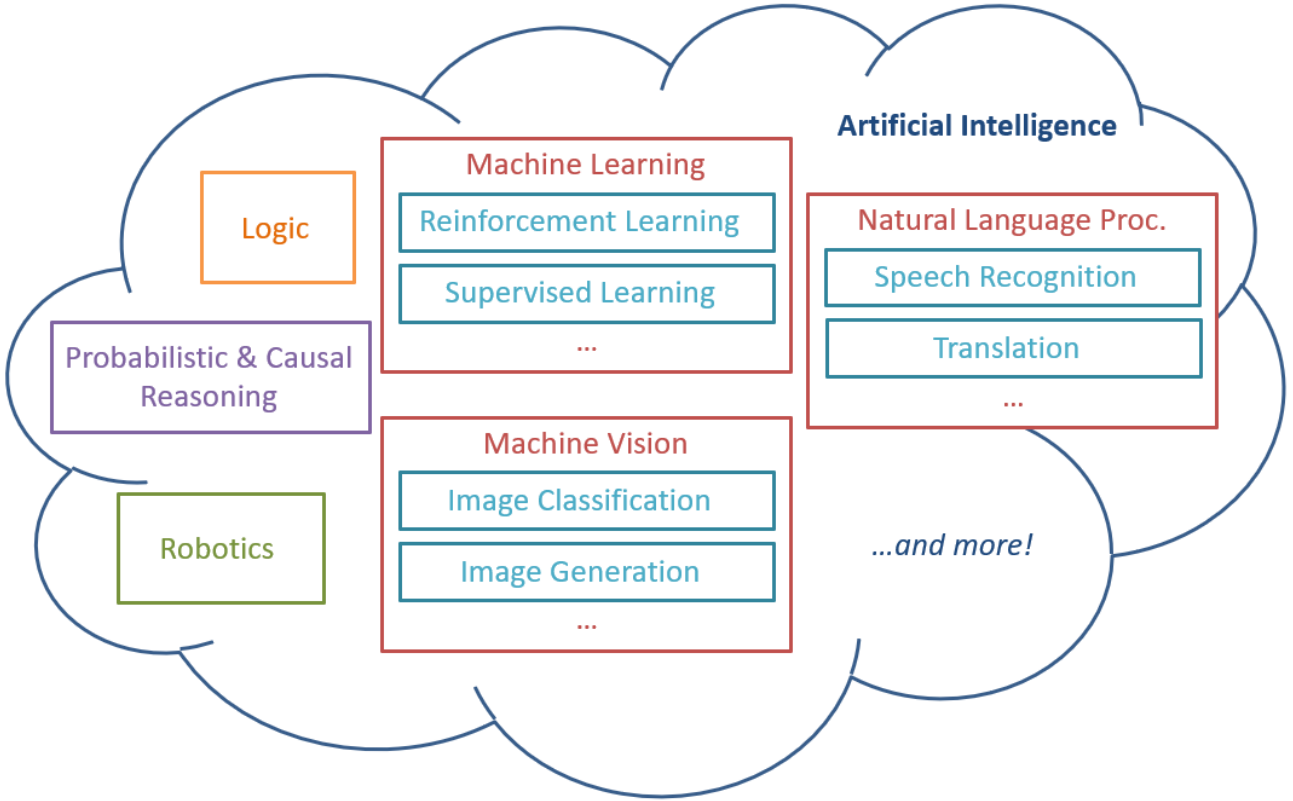
Artificial Specific Intelligence (ASI) refer to AI systems designed to perform **specific tasks** or **operate within a limited domain**.

Discussion



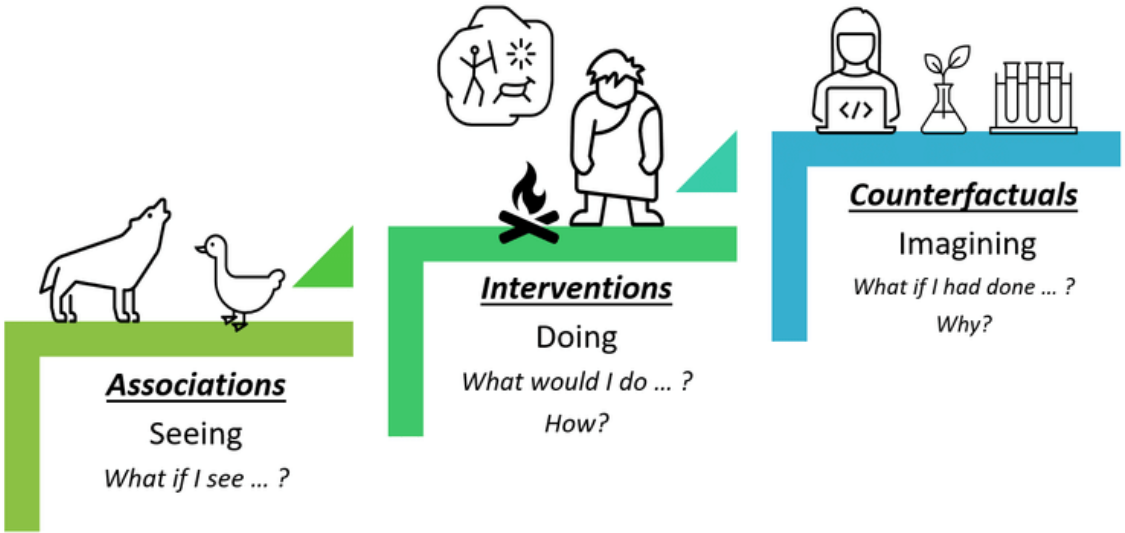
- What are some facets of human intelligence for which we have subfields of AI to study?
- For each of the subfields, what are the human intelligence facts you would NOT expect?

Types of Intelligence



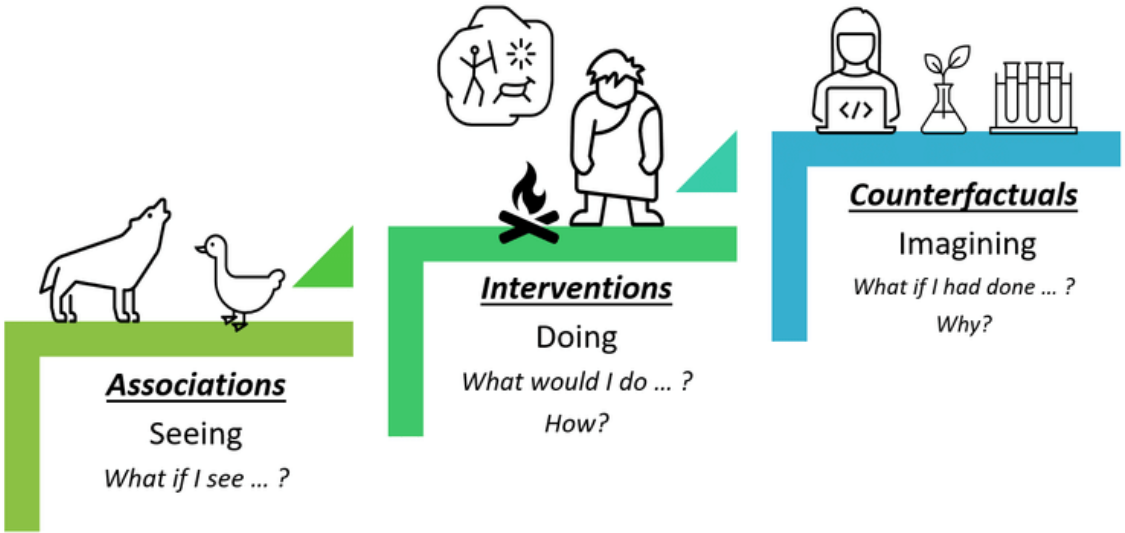
Levels of Intelligence

- **Associational** (L1): **Animalistic** intelligence in which correlations are drawn between variables. Often used for **prediction and imitation** tasks.



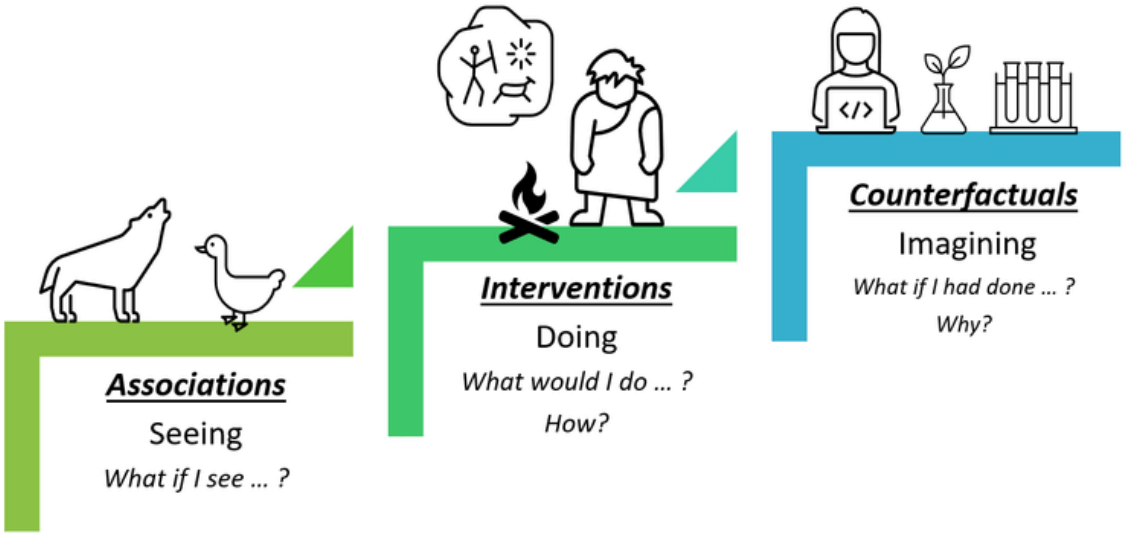
Levels of Intelligence

- **Interventional** (L2): **Causal** understanding of **what actions lead to what outcomes**, like a toddler learning to interact with its environment.



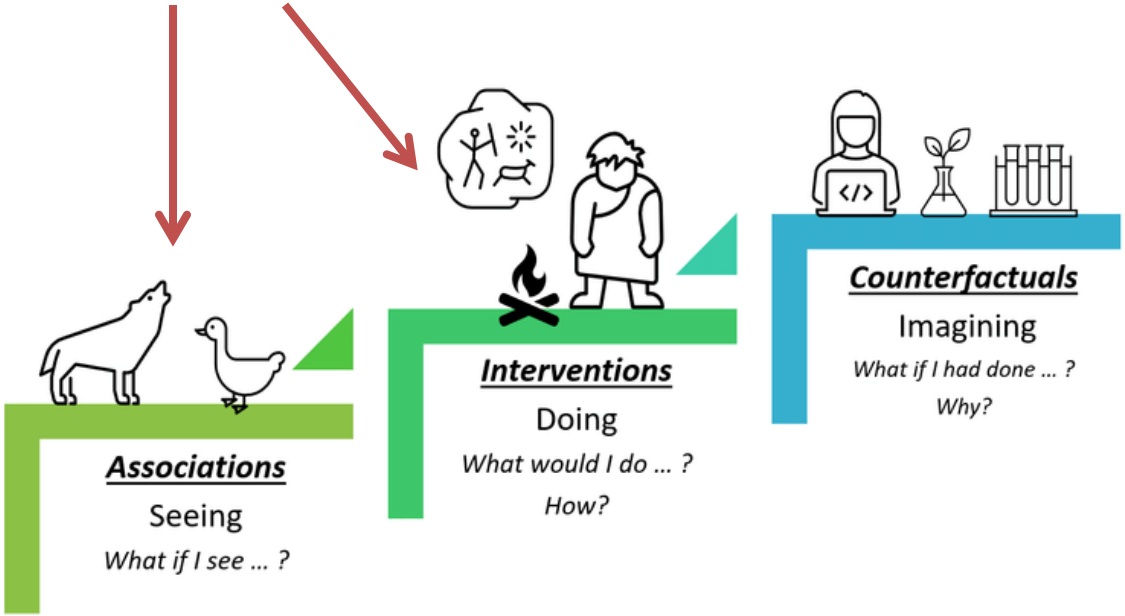
Levels of Intelligence

- **Counterfactual** (L3): **Human-level** understanding of systems of cause and effect: ability to make assertions about events that are contrary to what happened, thus decoupling cognition from data.



Levels of Intelligence

- Where is machine learning and modern AI?



This course

- Breadth coverage
- Practice meets theory
- NOT just machine learning
- NOT just deep learning
- NOT much of philosophy

Remaining of the lecture

CLASSWORK 1